

connecting BCDB. They join a body at B. However, in the real world, you'll see some mutant butterflies.

The various turns in the bullish butterfly are governed by Fibonacci ratios. Let's go through each one to show how I find them.

AB/AX retrace. I use the ratio of the various turns to see if the result is near a Fibonacci number. In these computations, I use the high–low value of the last point. You'll see what I mean.

The low price at X is 19.40, the high at A is 29.05, and the high–low range at B is 22.19 to 20.94. The AB/AX ratio becomes, using the low at B, $(29.05 - 20.94)/(29.05 - 19.40)$ or .84. Using the high at B, we get $(29.05 - 22.19)/(29.05 - 19.40)$ or .71. The span of .71 to .84 encompasses the .786 target Fibonacci number (from [Table 17.1](#)), so B is a proper turning point for the pattern.

CB/AB retrace. Similarly, I find the CB retrace of AB, using the high–low range of C (26.96 to 25.62) and the high at A, low at B. Plugging the high at C into the equation, we find $(26.96 - 20.94)/(29.05 - 20.94)$ or .742. Using the low at C, we get $(25.62 - 20.94)/(29.05 - 20.94)$ or .577. The range .577 to .742 must span at least one of the numbers listed in [Table 17.1](#). Two numbers qualify, .618 and .707, so turn ABC qualifies as valid.

CD/CB extension. I find the extension the same way, using the high–low range of the last point (point D in this case). D has a high price of 18.47 and a low of 17.01. Plugging in the numbers gives a span of 1.41 to 1.65. The 1.618 number squeezes into the range, so we found point D.

AD/AX retrace. Because Fibonacci-based patterns are so complicated, I use the high–low range as discussed to qualify the turn, except for AD/AX. For this ratio, I use a 3% window. I use the high at A, low at D, and low at X in the formula, which gives a value of 1.25, which is within 3% of 1.27. Thus, we have a valid bullish butterfly whose turns come close to the Fibonacci numbers listed in the table.

Volume. We'll have a report on volume later in the Statistics section, but volume trends downward most often as measured from the start of the pattern (X) to the end (D). That's in bear markets. The volume trend in bull markets is about random.

Duration. I limited patterns to 6 months or less, but that's an arbitrary limit.